# “SMARTCINNAMON”: A MULTI-MODAL MACHINE LEARNING AND IOT FRAMEWORK FOR PRECISION CINNAMON CULTIVATION

RP-25-26J-375

Project Proposal Report

Vihani W.W.G.D

B.Sc. (Hons) Degree in Information Technology

Specializing In Information Technology

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# DECLARATION

We declare that this is our work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date 28/08/2025

Signature of the Co - supervisor: Date

**ABSTRACT**

**Post-Harvest Ceylon Cinnamon Warehouse Quality Guard**

Ceylon cinnamon is a premium export crop for Sri Lanka, yet post-harvest quality is often compromised due to inadequate warehouse monitoring. Traditional methods track only temperature and humidity, failing to prevent mold, microbial contamination, and quality loss, which increases rejection risks under HACCP and Geographical Indication (GI) standards. This study proposes the Post-Harvest Cinnamon Quality Guard, an IoT and AI-based monitoring system. Multi-sensor arrays installed at warehouse points continuously measure multi environmental parameters: temperature, relative humidity, $CO_2$, $O_2$, VOCs, light intensity, airflow, dust/particle levels, vibration, and surface moisture. Data is transmitted to a cloud/mobile dashboard every few minutes, where machine learning models compare readings against predefined baselines. When risks are detected, the system issues real-time alerts and suggests corrective actions, such as adjusting ventilation, activating dehumidifiers, or reducing light exposure. Anticipated results include reduced mold development, fewer post-harvest losses, and consistent compliance with HACCP and GI requirements. The system is expected to improve product consistency, reduce export rejections, and strengthen Sri Lanka's cinnamon industry in global markets. In conclusion, the PostHarvest Cinnamon Quality Guard offers a cost-effective, proactive, and data-driven solution for cinnamon exporters, with potential to expand into other high-value spice industries.

**Keywords: Cinnamon quality, IoT monitoring, post-harvest management, HACCP compliance**

# TABLE OF CONTENTS

DECLARATION ..... 3
ABSTRACT ..... 4
LIST OF FIGURES ..... III
LIST OF TABLES ..... III
LIST OF ABREVATIONS ..... IV
1. INTRODUCTION ..... 1
1.1 BACKGROUND AND LITERATURE SURVEY ..... 2
2. RESEARCH GAPS ..... 4
3. RESEARCH PROBLEM ..... 7
4. OBJECTIVES ..... 9
4.1 Main Objectives ..... 9
4.2 Specific Objectives ..... 9
**4.2.1 Multi-Parameter Environmental Monitoring ..... 9**
**4.2.2 Baseline-Driven Data Analysis with Machine Learning ..... 9**
**4.2.3 Real-Time Alerts and Corrective Guidance ..... 9**
**4.2.4 Cloud-Based Dashboard and Visualization ..... 10**
5. METHODOLOGY ..... 11
5.1 System Overview Diagram ..... 11
5.2 Component System Diagram ..... 13
5.3 Project Technology Stacks ..... 14
5.4 Software Solution ..... 15
6.PROJECT REQURMENTS ..... 16
6.1 Feasibility Studies ..... 16
6.2 Survey Result ..... 16
6.3 User Requirements ..... 17
6.4 Functional Requirements ..... 17
6.5 Non-Functional Requirements ..... 17

6.6 Hardware Requirements ..... 18
6.7 Software Requirements ..... 18
**6.7.1 Backend Infrastructure ..... 18**
**6 7.2. Development Tools ..... 18**
6.8 Test Cases ..... 19
7. DESCRIPTION OF PERSONAL AND FACILITIES ..... 21
8. GANT CHART ..... 22
9. WORK BREAKDOWN CHART ..... 23
10. COMMERCIALIZATION PLAN ..... 24
10.1. Business Plan ..... 24
10.2 Marketing Plan ..... 25
11. BUDGET AND BUDGET JUSTIFICATION ..... 26
12. COCLUSION ..... 27
13. REFERENCES ..... 28
14. APPENDICIES ..... 30

## LIST OF FIGURES

Figure 1 : System Overview Diagram ..... 11
Figure 2 : Component System Diagram..... 13
Figure 3 : SDLC Life Cycle ..... 13
Figure 4 : Gant Chart ..... 15
Figure 5 : WBS Diagram ..... 22
Figure 6 : : Business Plan..... 24
Figure 7 : Appendices I ..... 28

## LIST OF TABLES

Table I : List of Abbreviations ..... IV
Table II : Research Gap ..... 6
Table III : Project Technology Stack ..... 14
Table IV : Test Cases ..... 19
Table V : Description of personal and facilities ..... 21
Table VI : Budget and Budget Justification ..... 26

# LIST OF ABREVATIONS

*Table I : List of Abbreviations*

| | |
|---|---|
| AI | Artificial Intelligence |
| BLE | Bluetooth Low Energy |
| MERN | MongoDB, Express, React, Node.js |
| CNN | Convolutional Neural Network |
| CSV | Comma-Separated Values |
| SDLC | Software Development Life Cycle |
| API | Application Programming Interface |
| IDE | Integrated Development Environment |
| LKR | Sri Lankan Rupee |
| DHT11 | Digital Humidity & Temperature sensor (Temp/RH) |
| WBS | Work Breakdown Structure |
| ML | Machine Learning |
| IoT | Internet of Things |
| MQTT | Message Queuing Telemetry Transport (IoT protocol) |
| RH | Relative Humidity |

# 1. INTRODUCTION

Cinnamon, particularly Ceylon cinnamon, is a high-value spice cherished for its quality and distinct aroma. However, its delicate nature makes it highly susceptible to environmental fluctuations during the post-harvest phase, potentially impacting its quality, aroma, and shelf life. Traditionally, monitoring these parameters involved basic temperature and humidity tracking, but such systems fall short in ensuring the optimal conditions required for preserving the cinnamon's premium grade [1].

To address these limitations, we have developed an innovative Cinnamon-Specific PostHarvest Monitoring System. This advanced system utilizes a multi-sensor array deployed at strategic points within cinnamon storage areas to continuously monitor a set of critical environmental parameters. These parameters include Temperature, Relative Humidity, $CO_2$ and $O_2$ levels, VOCs (Volatile Organic Compounds), Light intensity, Airflow speed, Dust levels, Vibration, and Surface moisture (condensation) [2].

The system not only tracks these factors in real-time but also employs machine learning models to compare live sensor data with pre-defined baseline thresholds for each parameter. By leveraging ML analytics, the system can proactively detect deviations that could lead to mold growth, degradation of aroma, or other quality risks. Real-time alerts are sent to a cloud-based dashboard or mobile application, providing farmers and warehouse managers with immediate corrective action guidance-whether it be adjusting ventilation, dehumidifying, or reducing light exposure [3] [4].

This proactive approach helps ensure cinnamon maintains its export-grade quality standards throughout storage, enhancing the overall efficiency and profitability of cinnamon production. By integrating advanced environmental monitoring with data-driven insights, this system sets a new standard in the post-harvest management of cinnamon, focusing on quality preservation, mold reduction, and aroma stability in a fully automated, intelligent manner [5].

## 1.1 BACKGROUND AND LITERATURE SURVEY

The post-harvest quality management of agricultural products, particularly high-value crops like Ceylon cinnamon, is a crucial aspect of maintaining their value. For cinnamon, the preservation of its distinctive aroma and prevention of mold growth are vital factors in maintaining its export grade quality. Traditional post-harvest management systems typically focus on basic parameters such as temperature and relative humidity. While these factors are essential, they do not fully address the complexity of environmental conditions that can affect the aroma and risk of mold in cinnamon storage.

Multi-Sensor Systems in Post-Harvest Management: Recent advances in multi-sensor systems have enabled more comprehensive monitoring of agricultural products, extending the scope of traditional monitoring systems. These sensors can track a wider array of environmental parameters such as $CO_2$ levels, VOC concentrations, light intensity, and surface moisture, in addition to the basic temperature and humidity. This enhanced monitoring provides deeper insight into the conditions that influence product quality [5]. For instance, [6] explored the use of multi-sensor systems in agricultural storage, highlighting how the integration of diverse environmental factors can significantly reduce risks like mold growth or aroma degradation. The Cinnamon-Specific Post-Harvest Monitoring System proposed in this study monitors ten environmental parameters that are critical to preserving cinnamon's quality, far beyond the traditional focus on temperature and relative humidity.

Machine Learning (ML) for Predictive Monitoring: One of the most significant advancements in agricultural monitoring is the use of machine learning (ML) to process the data collected by IoT sensors. ML models can analyze real-time data, comparing it against baseline thresholds for each parameter-such as a relative humidity range of 55–65% and temperature between 20–25°C. By detecting deviations from these thresholds, ML models can identify risks to the cinnamon's quality, such as mold growth or aroma degradation, even before they become visible. This predictive capability ensures that proactive measures can be taken to maintain the quality of cinnamon [7] [8] [9].

For example, and emphasize the growing role of ML in real-time monitoring, allowing for dynamic adjustments in environmental conditions based on ongoing analysis. By integrating these ML models into the post-harvest monitoring system, the proposed system can automatically adjust environmental conditions such as temperature, humidity, and airflow, thus preserving the export-grade quality of the cinnamon. Real-Time Corrective Actions and Alerts: The ability to provide real-time corrective actions is another critical feature of modern post-harvest monitoring systems. [4] underscores the importance of IoT-based alert systems that can send notifications for immediate interventions. These alerts can guide actions such as adjusting ventilation, activating dehumidifiers, or adjusting lighting to bring environmental parameters back into the desired ranges. This is particularly important for maintaining the long-term quality of stored cinnamon,

preventing issues such as mold development or loss of aroma. The Cinnamon-Specific Post-Harvest Monitoring System includes an alert system that not only detects deviations but also recommends corrective actions in real time. This feature ensures that any potential quality issues are addressed before they impact cinnamon's marketability.

Cloud and Mobile-Based Dashboards for Continuous Monitoring: The integration of cloud-based and mobile dashboards allows for continuous monitoring of environmental conditions, ensuring that stakeholders can access data remotely.  explores the role of cloud platforms in agricultural systems, enabling remote access to live data, which is especially useful for managers overseeing multiple storage locations. This system allows warehouse managers to monitor cinnamon storage conditions in real time, providing insights into long-term storage trends and ensuring consistent quality across locations [10]. These dashboards also support data-driven decision-making, empowering managers to take informed actions regarding the storage and preservation of cinnamon.

Community Engagement in Agricultural Systems: In addition to technological solutions, community engagement plays an essential role in the success of agricultural systems. Lia et al. [1] highlights the effectiveness of using text-messaging chatbots to engage communities in environmental monitoring. By integrating similar chatbot technology into the post-harvest monitoring system, the system can facilitate real-time communication between farmers, warehouse managers, and other stakeholders. This shared platform allows for better collaboration, helping managers make quicker decisions and share knowledge about cinnamon storage and quality management.

Policy and Sustainability Considerations: Land use and agricultural policies also play a role in the effective implementation of smart agriculture technologies. [3] examines how land use policies must be adaptable to changing agricultural practices, particularly when adopting technologies like IoT and ML. These technologies can optimize resource use and improve the sustainability of farming practices. When applied to the Cinnamon-Specific Post-Harvest Monitoring System, such policies would ensure that the system is aligned with environmental standards and can effectively contribute to the sustainability of cinnamon farming practices.

The integration of AI, IoT, and ML into post-harvest monitoring systems represents a significant breakthrough in agricultural technology. The Cinnamon-Specific Post-Harvest Monitoring System combines continuous environmental monitoring, real-time alerts, and predictive capabilities to ensure the cinnamon remains within optimal conditions for export. The system goes beyond traditional monitoring by using machine learning to proactively manage the environment and prevent risks to cinnamon's quality. Furthermore, the system's integration with cloud and mobile platforms enables remote access, facilitating better decision-making and operational efficiency. Ultimately, this data-driven, proactive approach not only preserves the aroma and quality of cinnamon but also enhances the overall efficiency and sustainability of the post-harvest process.

## 2. RESEARCH GAPS

This table and comparative analysis illustrate the key research gaps discovered by examining existing cinnamon post-harvest storage technologies and methodologies. Research A, B, C, and D represent different systems, each addressing certain aspects of monitoring and control but leaving important requirements unmet.

- **Research A (Tharaka et al., 2021)** This system primarily focuses on temperature and relative humidity (Temp/RH) control within a hybrid cinnamon chamber. It is capable of issuing Bluetooth alerts and even initiating basic automated responses such as activating fans or heaters when thresholds are exceeded. While this is useful for maintaining a stable microclimate, the system is limited in scope. It does not incorporate multi-parameter sensing (e.g., $CO_2$, $O_2$, VOCs, airflow, dust/particle levels, vibration, surface condensation), which are essential to accurately capture the complex conditions influencing cinnamon quality. Moreover, it lacks any machine learning (ML)–based risk prediction or prescriptive decision-support, leaving operators without proactive guidance to prevent mold growth or aroma loss.

- **Research B (Jarašūnienė et al., 2023)** This system emphasizes IoT connectivity and cloud-based monitoring, offering Wi-Fi dashboards and real-time alerting across warehouse environments. While it demonstrates scalability and remote accessibility, the solution is generic and not tailored to cinnamon-specific requirements. It lacks specialized sensing for aroma-related VOCs, gas concentrations, and surface moisture, which are vital to detect early signs of quality degradation. Furthermore, spatial distribution of sensors (door, center, corners) is not fully addressed, resulting in potential blind spots. Importantly, the system provides only alerts without ML-driven risk prediction or prescriptive guidance to help operators take corrective actions.

- **Research C (Sathya et al., 2023)** This IoT-based warehouse monitoring and alerting prototype integrates multi-parameter sensing (temperature, relative humidity, gas detection) and demonstrates multi-node deployment, making it more advanced than earlier single-node systems. It provides rule-based threshold alerts through app/SMS notifications, ensuring timely awareness of deviations. However, the approach remains reactive and not cinnamon-specific, as it does not include baseline thresholds optimized for cinnamon storage. Additionally, the system lacks machine learning models that can anticipate mold/aroma risks before they occur, and it does not offer detailed prescriptive actions to guide operators in decision-making.

- **Research D (Feltes et al., 2023)** This research focuses on volatile organic compounds (VOCs) for differentiating true and false cinnamon, applying advanced sensing and machine learning techniques for classification. While valuable for cinnamon authenticity

and aroma studies, the system does not target the post-harvest warehouse environment or mold/aroma risk management. It omits critical aspects such as distributed IoT sensing, environmental monitoring (Temp/RH, airflow, surface condensation), and real-time alerts. Consequently, it provides useful domain-level insights into cinnamon quality but cannot serve as a comprehensive warehouse monitoring and risk-prevention solution.

*Table 2.1 : Research Gap*

| Conducted By | Multi-parameter sensing | Cinnamon-specific storage baselines | ML risk prediction | Real-time alerts & actions | Warehouse-wide deployment (multi-node) |
|---|---|---|---|---|---|
| **ResearchA Tharaka et al. (2021) (SLJoT)[11]** | ✗ (Temp/RH only) | ✓ (Temp/RH limits) | ✗ | ✓ (Bluetooth alerts; auto fan/heater) | ✗ (single chamber) |
| **ResearchB Jarašūnienė et al. (2023) (Sensors)[12]** | ✗ | ✗ | ✗ | ✗ | ✓ |
| **ResearchC IoT-Based Intelligent Warehouse Monitoring & Alerting (2023)[3]** | ✓ (Temp/RH + gas; threshold rules) | ✗ | ✗ | ✓ (app/SMS threshold alerts) | ✓ (multi-node prototype) |
| **Research D Feltes et al. (2023) (Micromachines)[5]** | ✓ (VOCs focus) | ✗ | ✓ (not warehouse risk) | ✗ | ✗ |
| **Proposed system (this research)** | ✓ (multi parameters; most effective selected) | ✓ (cinnamon baselines) | ✓ (risk score + feature rules) | ✓ (app/SMS + optional auto-control) | ✓ |

**Proposed System Contribution**

The suggested "Cinnamon Quality Guard" system is designed to bridge these gaps by:

- Monitoring 10 critical environmental parameters (Temp, RH, $CO_2$, $O_2$, VOCs, airflow, dust, light, vibration, surface moisture).
- Using cinnamon-specific baselines for decision rules.
- Applying ML models to calculate a risk score and predict mold/aroma risks early.
- Providing prescriptive alerts (e.g., activate dehumidifier, increase ventilation, reduce light exposure).
- Supporting multi-node warehouse deployment with cloud/mobile dashboards for full traceability.

This comprehensive approach ensures a proactive, data-driven safeguard for Ceylon cinnamon, maintaining export-grade quality.

## 3. RESEARCH PROBLEM

Ceylon cinnamon, also called "true cinnamon," is one of the most valuable spices in the world, and Sri Lanka is the leading producer and exporter recognized for unique aroma, flavor, and health benefits [1]. Cinnamon plays an important role in the Sri Lankan economy and supports many livelihoods; maintaining post-harvest quality is crucial to preserve value, satisfy customers, and protect the country's reputation in international markets. However, keeping cinnamon high quality during storage remains a significant challenge for the industry [13].

After harvest, cinnamon is typically stored in warehouses before processing, packaging, and exporting, during which it is highly sensitive to environmental conditions. Even small changes in temperature, humidity, airflow, or exposure to dust and contaminants can affect quality; high humidity or condensation encourages mold, while poor ventilation or exposure to light and chemical gases can reduce aroma-the key feature that distinguishes Ceylon cinnamon from cassia [5] [14] [15]. Aroma loss and mold contamination reduce market value, cause financial losses, and risk Sri Lanka's premium reputation; therefore, controlling the storage environment is vital to prevent spoilage and retain export standards.

Currently, warehouse monitoring is basic-most sites log only temperature and relative humidity or rely on manual inspections. While these two parameters matter, they are insufficient to fully protect quality; other influential factors include oxygen ($O_2$), carbon dioxide ($CO_2$), volatile organic compounds (VOCs), airflow speed, light intensity, dust/particle accumulation, vibration/shocks, and surface moisture/condensation. Ignoring these factors enables mold formation, aroma degradation, and other quality issues; the absence of a comprehensive system lets risks go unnoticed until corrective action is too late.

Another major limitation is that existing methods are reactive rather than proactive. Many systems only record data or alert after limits are exceeded (e.g., humidity spikes), by which time mold or aroma loss may already be underway [4] [11]. Reactive designs prevent managers from anticipating problems and taking preventive action, leading to losses in quality and quantity; for a high-value export commodity like Ceylon cinnamon, this is a serious concern for income, reputation, and long-term sustainability [3].

To address these problems, a dedicated cinnamon-specific post-harvest monitoring system is needed. Such a system should monitor multiple parameters in real time, not just temperature and humidity, with sensors deployed at corners, the center, and entry points to capture representative data. It should measure temperature, relative humidity, $CO_2$, $O_2$, VOCs, airflow, light intensity, dust/particles, vibration/shocks, and surface moisture, to build a complete picture of storage conditions and identify areas at risk of deterioration [2] [10].

In addition, the system should apply machine-learning (ML) analytics to incoming sensor data, comparing readings to predefined baselines for optimal storage and detecting deviations that signal risk [3] [5] [8]. This proactive approach enables early warnings and corrective actions before

quality is affected-for example, suggesting dehumidification or improved ventilation when rising humidity coincides with low airflow, or mitigating light/VOC exposure when thresholds trend upward. A cloud-based dashboard should provide real-time visibility and trend tracking to support informed decisions and audit-ready records [4].

In summary, the industry lacks an advanced, comprehensive, and proactive monitoring system tailored to cinnamon. Current methods are limited to basic parameters and reactive alerts, failing to address the multi-factor environmental risks that drive mold and aroma loss-leading to reduced export quality, economic losses, and reputational damage. A cinnamon-specific "quality guard" that integrates multi-sensor monitoring with ML analytics can detect risks early, provide prescriptive alerts, and visualize long-term trends-helping preserve Ceylon cinnamon's unique aroma and quality, reduce post-harvest losses, support farmers and exporters, and strengthen Sri Lanka's position in the global spice market.

# 4. OBJECTIVES

## 4.1 Main Objectives

SMARTCINNAMON is an integrated AI, IOT, GIS, and machine learning based system that enhances cinnamon production, post-harvest quality, and disease management. It includes real time soil nutrient monitoring with precision fertilizer recommendations, post-harvest environmental monitoring for mold/aroma control with audit-ready logs, GIS-based field utilization and yield forecasting for optimized replanting, and mobile-based fungal disease detection with severity and spread estimation improving yield, quality, compliance, and sustainability across the cinnamon value chain.

## 4.2 Specific Objectives

### 4.2.1 Multi-Parameter Environmental Monitoring

One of the primary objectives of this research is to design and implement a system that continuously monitors multiple environmental factors inside cinnamon storage warehouses. Unlike conventional systems that track only temperature and relative humidity (RH), this solution integrates a multi-sensor array capable of measuring ten critical parameters: Temperature, RH, $CO_2$, $O_2$, VOCs, light intensity, airflow speed, dust/particle levels, vibration/shock, and surface moisture. By capturing these parameters every few minutes, the system provides a comprehensive view of warehouse conditions, ensuring early identification of risks that could compromise cinnamon quality.

### 4.2.2 Baseline-Driven Data Analysis with Machine Learning

The second objective is to integrate baseline thresholds and ML-driven analytics for proactive risk detection. Each parameter will be compared against scientifically defined baselines (e.g., RH 55-65%, Temperature 20–25°C, VOC < threshold). The system will apply machine learning models to analyze deviations, detect abnormal patterns, and predict potential risks of mold growth, microbial activity, or aroma degradation. This goes beyond simple monitoring, enabling the system to function as a predictive quality guard rather than a reactive tool.

### 4.2.3 Real-Time Alerts and Corrective Guidance

Another objective is to provide real-time alerts with actionable recommendations whenever environmental parameters exceed safe baselines. Alerts will be delivered via mobile and web

dashboards, along with corrective suggestions such as activating dehumidifiers, improving airflow, or reducing light exposure. By combining monitoring with decision support, the system helps warehouse staff take timely actions that minimize post-harvest losses and preserve export-grade quality.

#### 4.2.4 Cloud-Based Dashboard and Visualization

To enhance usability, the system will include a cloud/mobile dashboard that visualizes real-time trends, historical data, and parameter correlations. This feature allows exporters and warehouse managers to easily track storage performance, identify recurring risks, and improve long-term storage strategies. The dashboard also strengthens HACCP and GI compliance by maintaining continuous digital records of storage conditions, ensuring audit readiness.

# 5. METHODOLOGY

## 5.1 System Overview Diagram

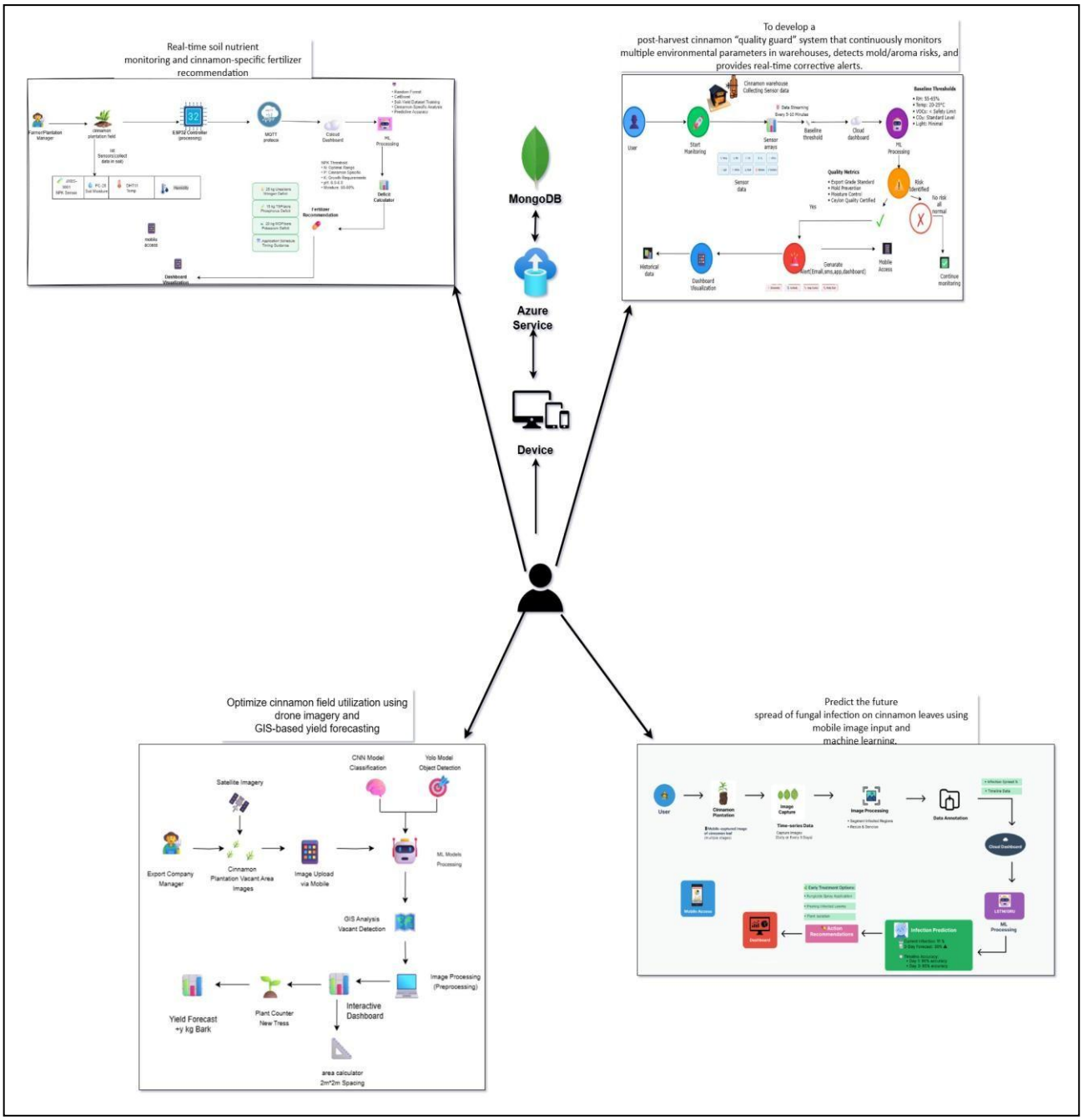


*Figure 5.1 : System Overview Diagram*

SMARTCINNAMON is an integrated AI–IoT–GIS platform for the cinnamon value chain. Field and warehouse devices stream data to Azure services and MongoDB; machine-learning models turn that data into real-time recommendations, risk alerts, forecasts, and dashboards delivered on mobile and web.
At the center is the user's device (mobile/web) connected to cloud services. IoT sensors in fields and warehouses send readings to Azure (via MQTT/REST) and are stored in MongoDB. Cloud ML services analyze the incoming data continuously and return insights to dashboards and alerting channels. The platform combines four modules that work together to raise yield, protect postharvest quality, and control disease.

**Real-time Soil & Fertilizer Intelligence -** ESP32 nodes read NPK (JXBS-3001), soil moisture (FC-28), and temperature/humidity (DHT11). Live values are compared with cinnamon-specific target ranges. ML models (e.g., Random Forest, Cat Boost) learn from soil + yield history to recommend exact fertilizer doses (e.g., kg/acre), showing a percent "deficit" for N, P, and K and converting that into actionable inputs.

**Post-harvest "Quality Guard" -** Multi-sensor stations monitor 10 factors inside warehouses: Temperature, Relative Humidity, $CO_2$, $O_2$, VOCs, light intensity, airflow, dust/particles, vibration/shock, and surface moisture. Baselines (e.g., RH 55-65%, 20-25 °C) and ML analytics detect mold/aroma risk early, trigger corrective guidance (ventilate, dehumidify, reduce light), and keep audit-ready logs and trends.

**GIS Field Planner & Yield Forecast -** Drone/satellite imagery is segmented to map **cultivated vs. vacant** areas. The system proposes optimal replanting layouts using standard spacing (e.g., 2 m × 2 m), estimates additional plant counts, and combines tree counts with historical bark productivity to forecast yield. Results appear on an interactive spatial dashboard: "X new trees → +Y kg expected yield."

**Leaf Disease Detection & Spread Prediction -** A mobile photo of a cinnamon leaf is analyzed by a CNN to detect current infection and severity. Time-series models (LSTM/GRU), trained on annotated progressions, predict how the infection will grow over the next few days (e.g., "+18% in 72 h"). The app visualizes the forecast and recommends early actions such as pruning, targeted spraying, or isolating plants.

**Data Flow -** Sensors/Images → Azure ingest → MongoDB storage → ML analytics → Dashboards & Alerts → Farm/warehouse actions.

**Value -** The platform delivers precise fertilizer dosing, proactive post-harvest quality control, optimized land utilization with yield gains, and early disease intervention-improving productivity, compliance, and sustainability across the cinnamon supply chain

## 5.2 Component System Diagram

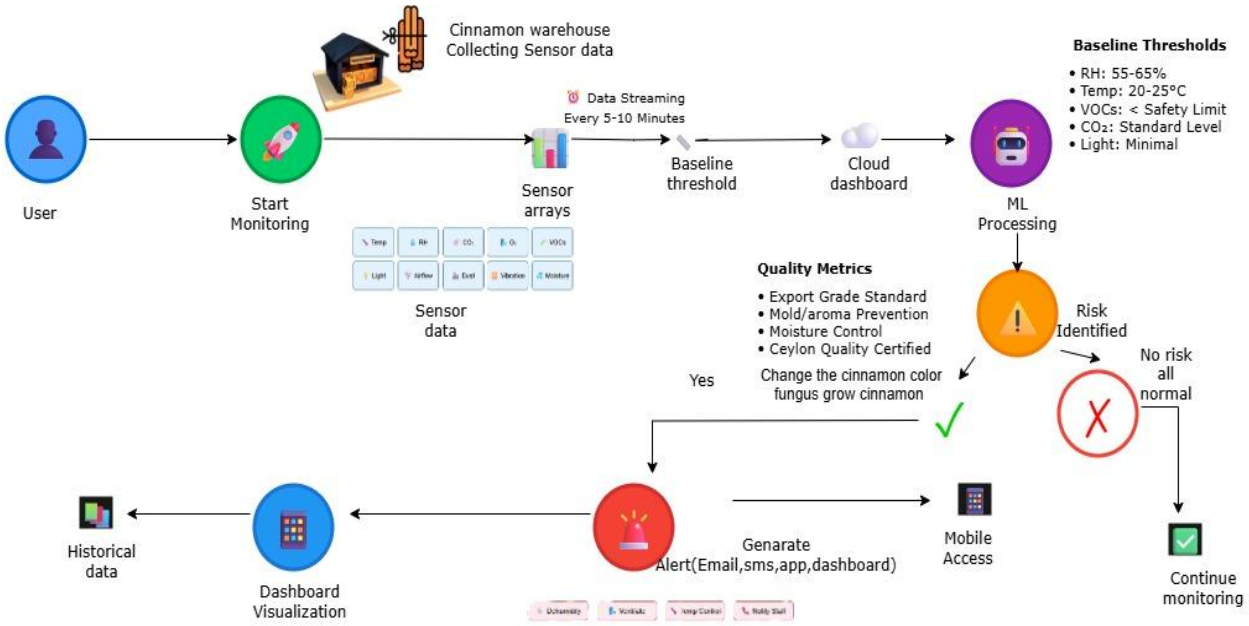


*Figure 5.2.1 : Component System Diagram*

According to the above system diagram, warehouse managers can ensure the safe storage of cinnamon by continuously monitoring multiple environmental parameters through IoT-based sensor arrays. The system captures real-time data on Temperature, Relative Humidity (RH), $CO_2$, $O_2$, VOCs, Light Intensity, Airflow Speed, Dust/Particles, Vibration/Shock, and Surface Moisture (condensation) at strategic points in the warehouse.

The captured data is streamed every 5-10 minutes to a cloud dashboard, where baseline thresholds and machine learning analytics are applied to identify risk conditions. If deviations are detected such as humidity levels exceeding 65% or temperature rising above safe storage limits-the system automatically triggers alerts. Corrective guidance is also provided in real time, such as activating dehumidifiers, improving ventilation, or adjusting airflow.

Unlike traditional monitoring systems that only track temperature and RH, this solution integrates 10 baseline-driven environmental parameters to provide a comprehensive assessment of storage conditions. The cloud/mobile dashboard displays storage health trends, alerts, and predictive risk visualizations, allowing managers to act before mold develops or aroma quality is lost.
By combining IoT sensing, AI-driven analytics, and real-time corrective action guidance, the PostHarvest Cinnamon Quality Guard minimizes post-harvest losses, safeguards Ceylon cinnamon's premium export quality, and ensures compliance with HACCP and GI standards. This proactive and data-driven approach provides exporters with a competitive edge in international spice markets

### 5.3 Project Technology Stacks

*Table 5.3.1: Project Technology Stack*

| Tools & Technologies | Purpose |
|---|---|
| React-Native | For mobile application frontend implementations. |
| React | For web application frontend implementations. |
| NodeJS | For backend implementation |
| Python | Machine learning-related implementations. |
| Django REST Framework | Machine learning-related implementations. |
| MongoDB | Use As Database |
| TensorFlow | Train machine learning models |
| IOT Sensors | For monitoring Temp, RH, $CO_2$, $O_2$, VOCs, airflow, dust, vibration, surface moisture. |
| ESP32 Microcontroller | For reading IoT sensor data and transmitting it to the server via MQTT. |
| MQTT Protocol | Lightweight protocol for real-time data transfer between sensors and cloud dashboard. |
| Visual Studio Code | Used as IDE for coding, debugging, and integration. |
| Git & GitHub | For version control and team collaboration. |
| Postman | For testing APIs and verifying IoT-cloud integrations. |

| | |
|---|---|
| Azure / AWS IoT Core | For cloud data storage, device management, and real-time monitoring platform. |
| Expo | For creating the mobile application |

### 5.4 Software Solution

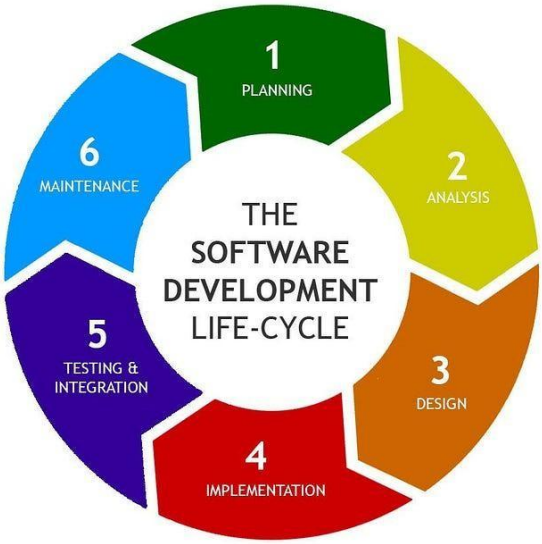


*Figure 5.4.2 : SDLC Life Cycle*

We will follow the Agile Software Development Life Cycle (SDLC) model to guide our project. Agile provides a flexible and iterative approach, allowing us to plan, develop, test, and deliver high-quality software in stages. This methodology encourages continuous improvement, frequent user feedback, and quick adjustments, ensuring the final product meets user needs effectively. By breaking the project into manageable sprints, the team can focus on delivering specific features incrementally, leading to a more efficient and responsive development process. This approach will help us build a robust and user-friendly land management application.

# 6.PROJECT REQURMENTS

## 6.1 Feasibility Studies

Conducting a feasibility study is essential to evaluate the potential for developing the proposed Post-Harvest Cinnamon Quality Guard system. This study examines the project from four perspectives: technical, operational, economic, and schedule feasibility.

**Technical Feasibility:** The system is technically achievable using reliable and well documented technologies. It integrates IoT multi-sensor arrays (for temperature, RH, $CO_2$, $O_2$, VOCs, light, airflow, dust, vibration, and surface moisture), ESP32 microcontrollers, and MQTT for data transmission. Machine learning models will be implemented using TensorFlow and scikit-learn for anomaly detection, while a cloud-based dashboard will be developed using MERN (MongoDB, Express, React, Node.js) or equivalent frameworks. The development team possesses the required expertise in IoT integration, AI/ML model development, and fullstack web/mobile application design, ensuring successful implementation.

**Operational Feasibility:** The system is designed to address the critical post-harvest challenges faced by exporters and warehouse operators. By providing real-time monitoring and early risk detection, it minimizes mold development, reduces post-harvest losses, and ensures HACCP/GI compliance. The interface will be simple and intuitive, with real-time alerts and corrective guidance accessible via mobile and web platforms. Early pilot testing with cinnamon warehouses will provide valuable feedback for refining the system's usability and effectiveness.

**Economic Feasibility:** The proposed solution is economically viable as it reduces losses caused by spoilage, rejections, and non-compliance in international markets. A subscription based model can be adopted, offering affordable access for warehouse operators and exporters (e.g., a monthly fee scaled by warehouse size). The long-term savings from reduced product rejection, improved quality assurance, and higher export consistency outweigh the development and operational costs, making the system cost-effective and sustainable.

**Schedule Feasibility:** The project will follow an Agile SDLC approach, enabling incremental development and flexibility. Core modules (sensor integration, baseline monitoring, and dashboard) will be delivered in the early phases, while advanced features such as ML-based anomaly detection and predictive analytics will be introduced iteratively. The system is planned for release within 12 months, aligning with the seasonal export cycles of Ceylon cinnamon, ensuring timely field adoption.

## 6.2 Survey Result

Data was collected through a survey distributed to cinnamon warehouse operators and exporters using Google Forms. The purpose of the survey was to understand the current practices, challenges, and needs related to post-harvest cinnamon storage and quality management. The survey included questions about storage preferences, environmental

monitoring practices, compliance with HACCP/GI requirements, and the usability of existing monitoring tools.

Responses were categorized and analyzed using statistical methods, pie charts, and bar graphs to visualize key findings. The analysis revealed common issues and needs, such as difficulties in continuous multi-parameter monitoring, dependency on manual inspections, and the absence of automated alert systems for mold and aroma risks. Many respondents also emphasized the lack of digital compliance records, which complicates audit readiness and increases rejection risks in export markets.

These insights were instrumental in shaping the development of our Post-Harvest Cinnamon Quality Guard system to address real-world challenges effectively by providing continuous monitoring, automated alerts, and data-driven quality assurance.

### 6.3 User Requirements

- Enable continuous monitoring of multiple environmental parameters (e.g., temperature, RH, $CO_2$, $O_2$, VOCs, light, airflow, dust, vibration, surface moisture).
- Provide real-time alerts with corrective guidance when parameters exceed safe thresholds.
- Offer a cloud/mobile dashboard for visualizing trends, patterns, and storage performance.
- Ensure HACCP and GI compliance support by maintaining baseline-driven monitoring.
- Provide accessibility for non-technical warehouse staff through a simple, user-friendly interface.

### 6.4 Functional Requirements

- Collect real-time readings from multi-sensor arrays installed at warehouse points.
- Compare sensor data against predefined baselines using ML-based analytics.
- Generate automated alerts with corrective suggestions (e.g., activate dehumidifiers, improve ventilation, reduce light exposure).
- Display trends, storage history, and risk predictions via mobile and web dashboards.
- Provide parameter-based reports to support compliance audits and decision-making.
- Allow warehouse staff to acknowledge alerts confirm corrective actions through the system.

### 6.5 Non-Functional Requirements

- **Usability:** The system must be intuitive and simple so that warehouse operators with limited technical knowledge can easily use it.
- **Reliability:** The system should deliver continuous monitoring with minimal downtime, ensuring accurate and uninterrupted data collection.
- **Performance:** Sensor readings and alerts must be processed in near real-time, providing accurate and timely insights.

- **Scalability:** The system should support integration of multiple warehouses and large sensor networks.
- **Security:** All data, including warehouse records and sensor readings, must be encrypted and securely stored.
- **Compatibility:** The system should be compatible with various devices and platforms, including Android, iOS, Windows, and macOS, ensuring that users can access the application from any device.

### 6.6 Hardware Requirements

- IoT Sensor Modules: Temperature, RH, $CO_2$, $O_2$, VOC, light, airflow, dust/particle, vibration, and surface moisture sensors.
- Microcontroller: ESP32 or Raspberry Pi for sensor integration and data transmission.
- Gateway Device: Wi-Fi-enabled hub for MQTT data streaming.
- Laptop/Desktop for dashboard access with stable internet connection.
- Mobile Devices (Android/iOS) for monitoring via mobile application

### 6.7 Software Requirements

- Mobile Operating System: The application should be compatible with the latest versions of iOS and Android.
- Desktop/Laptop: Windows 10/11, macOS

#### 6.7.1 Backend Infrastructure

• Database: MongoDB for storing sensor readings, alerts, and historical records.
• Backend Framework: Node.js for handling server-side logic and APIs.
• Machine Learning Libraries: TensorFlow for ML-based anomaly detection and prediction.
Cloud Integration: Firebase/AWS IoT Core for real-time data management and notifications.

#### 6 7.2. Development Tools

- IDEs: Visual Studio Code for coding and debugging.
- Version Control: Git & GitHub for collaborative development and code management.
- API Platform: Flask / FastAPI for developing API services and ML model integration.
- Dashboard & Visualization: React.js (web) and React Native (mobile) for user interfaces.
- Data Analytics Tools: Pandas, NumPy, Matplotlib for model development and testing.

## 6.8 Test Cases

*Table 6.8.1 : Test Cases*

| Test Scenario | **Monitoring Multi-Sensor Data** |
|---|---|
| Test Scenario ID | **TC1** |

| Description | Pre - Conditions | Inputs | Expected Results | Post-conditions |
|---|---|---|---|---|
| Verify if the system can read and record data from all 10 environmental sensors. | • Sensors (Temperature, RH, $CO_2$, $O_2$, VOCs, Light, Airflow, Dust, Vibration, Surface Moisture) must be connected.<br>• Central dashboard must be active. | Live sensor readings streamed every 5–10 minutes. | Dashboard displays accurate realtime readings for all parameters. | Should be able data is logged for trend analysis and future reference. |

*Table 6.8.2: Test Cases*

| Test Scenario | **Detecting Humidity Risk** |
|---|---|
| Test Scenario ID | **TC2** |

| Description | Pre - Conditions | Inputs | Expected Results | Post-conditions |
|---|---|---|---|---|
| Check if the system identifies high RH risk that could cause mold. | • RH sensor installed and calibrated.<br>• Baseline set at 55–65% RH. | RH value rises above 70%. | System triggers "High Humidity Alert" with corrective action (e.g., activate dehumidifier). | Alert is recorded and user action is updated in system logs. |

*Table 6.8.3: Test Cases*

| Test Scenario | **Detecting Temperature Deviation** |
|---|---|
| Test Scenario ID | **TC3** |

| Description | Pre - Conditions | Inputs | Expected Results | Post-conditions |
|---|---|---|---|---|
| Ensure the system flags abnormal temperature changes. | • Temperature sensor active and calibrated. • Baseline set at 20–25°C. | Input temperature = 30°C. | Alert generated: "Temperature exceeds safe range – adjust ventilation." | Data saved, deviation marked in system report. |

*Table 6.8.4: Test Cases*

| Test Scenario | **Dashboard Visualization** |
|---|---|
| Test Scenario ID | **TC4** |

| Description | Pre - Conditions | Inputs | Expected Results | Post-conditions |
|---|---|---|---|---|
| Verify if the dashboard displays historical and real-time data trends. | • Sensors transmitting data properly. • User logged into system. | User selects "View Trends" option. | Dashboard shows graphs of past 24-hour sensor data (Temp, RH, $CO_2$, etc.). | Data visualization saved for reports and export. |

# 7. DESCRIPTION OF PERSONAL AND FACILITIES

*Table 7.1 : Description of personal and facilities*

| Registration No | Name | Task Description |
|---|---|---|
| IT22594372 | Vihani W.W.G. D | • Analyze environmental factors affecting post-harvest cinnamon quality (temperature, RH, $CO_2$, $O_2$, VOCs, etc.).<br>• Collect requirements from warehouse operators and exporters<br>• Implement multi-sensor integration and baseline threshold configuration.<br>• Develop cloud/mobile dashboard for real-time data visualization.<br>• Design and test ML-based alert system for mold and quality risk detection.<br>• Evaluate accuracy of alerts and fine-tune sensor calibration.<br>• Implement historical data storage and trend analysis features. |

# 8. GANT CHART

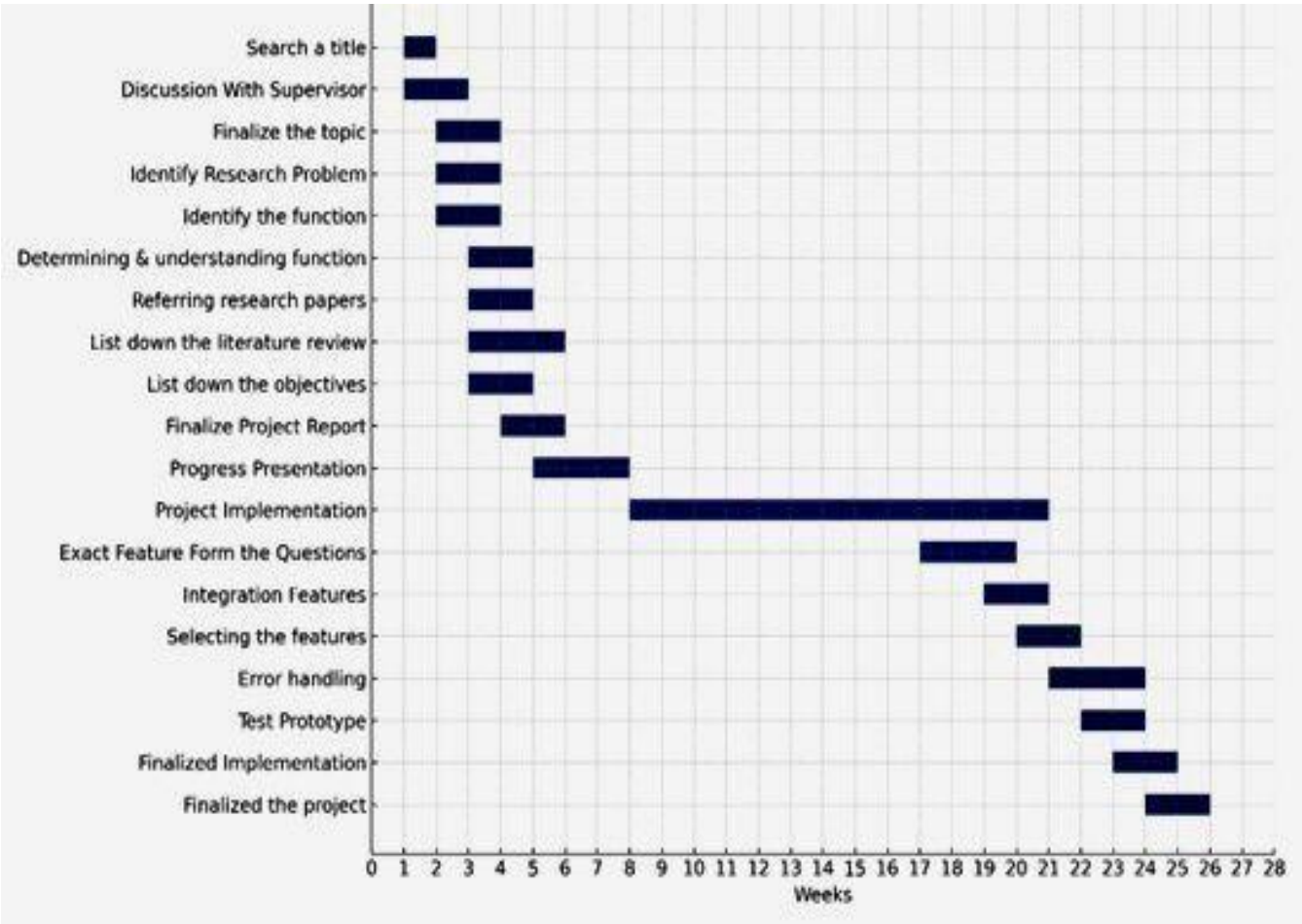


*Figure 8.1: Gant Chart*

# 9. WORK BREAKDOWN CHART

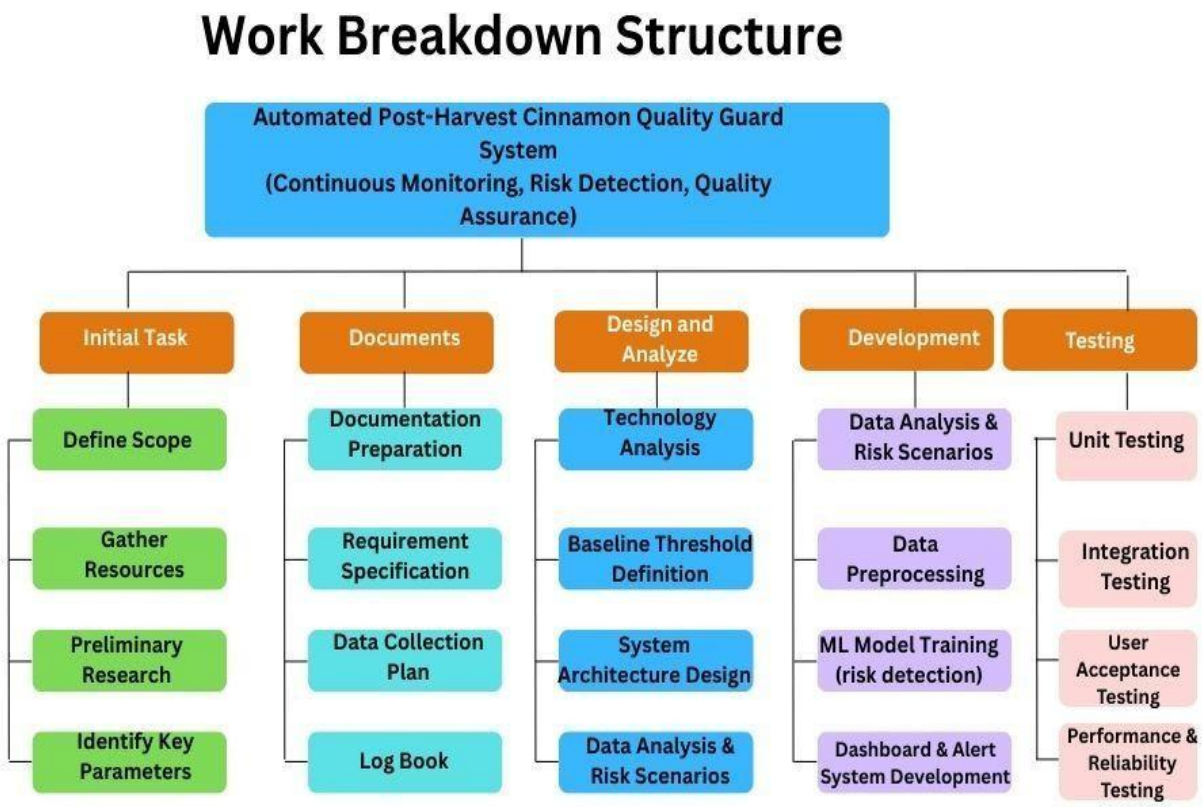


*Figure 9.1: WBS Diagram*

# 10. COMMERCIALIZATION PLAN

## 10.1. Business Plan

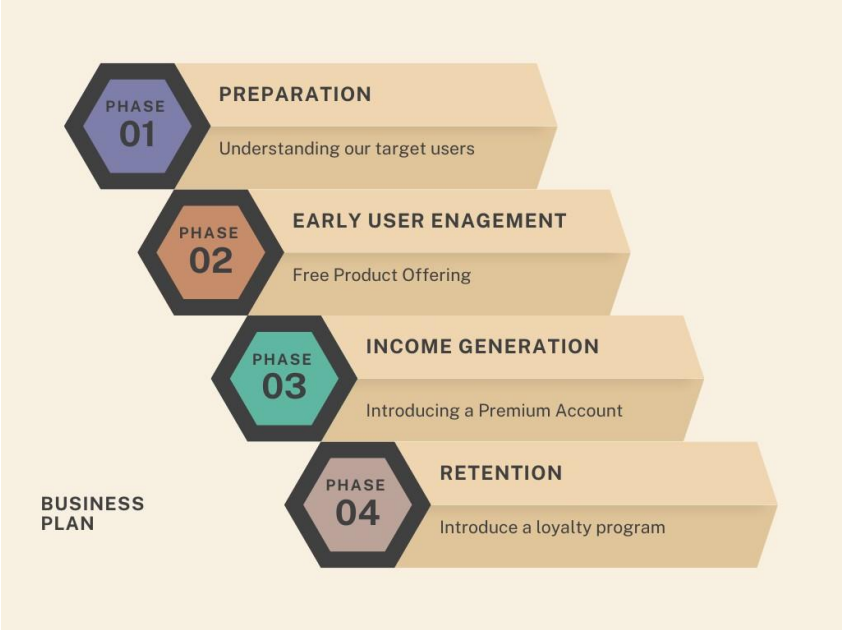


*Figure 10.1: Business Plan*

**Phase 01 -** In this phase, we will prepare everything needed before launching the product. This includes understanding our target users and making sure the application works perfectly.

**Phase 02 -** We will offer the product for free to early users, especially in our target market. This is important to get feedback from farmers and land managers about how well the app works for them. Their feedback will help us make the app even better.

**Phase 03** - After improving the app based on feedback, we will begin charging for some advanced features. These might include detailed financial tools and extra storage options. The basic paid plan will start at 400 LKR per month, which is affordable for most small and medium-sized farms. We hope to encourage users who tried the free version to upgrade to a paid plan for added benefits.

**Phase 04** - In this final phase, we will focus on keeping our customers happy and encouraging them to stay. We will phase out the free version and introduce a loyalty program where users

earn points for sticking with the service or referring to others. These points can be used for discounts or extra features, helping to build long-term customer loyalty.

### 10.2 Marketing Plan

**Social media marketing -**We will use social media to spread the word about our app. Our goal is to make people aware of the app, get them interested, and encourage them to subscribe. We'll create simple, helpful content like how-to videos, stories from real users, and tips on using the app. We'll focus on platforms that our target audience uses, like Facebook, Instagram, and WhatsApp.

**Sales approach**-Our team will reach out directly to small and medium-sized farms. We'll do this through farming groups, local events, and online webinars where we can show how the app works and how it can help solve common farming problems.

**Creating customer package-**We will offer different subscription plans, from basic to premium. Each plan will have different features, so users can choose what best fits their needs and budget. We'll encourage users to move from the free plan to a paid plan by offering more value with the higher-tier packages

# 11. BUDGET AND BUDGET JUSTIFICATION

*Table 11.1:Budget and Budget Justification*

| Description | Amount (LKR) |
|---|---|
| IoT Sensors (NPK JXBS-3001, Moisture FC-28, Temp/Humidity DHT11, pH/EC Probes) | 40,000 |
| ESP32 Microcontrollers + Wi-Fi/LoRa Modules | 15,000 |
| Cloud Hosting & APIs (Azure, MQTT broker, Firebase/MongoDB Atlas) | 25,000 |
| ML/AI Model Training Infrastructure (Cloud GPU credits / Colab Pro / Azure ML) | 20,000 |
| Drone/Satellite Imagery (GIS Yield Planner) | 10,000 |
| Mobile Application Development (Fungal Spread Predictor) | Free |
| Multi-Sensor Array for Post-Harvest Quality Guard (Temp, RH, $CO_2$, VOC, Light, Airflow quality , motion, Surface Moisture) | 10,000 |
| Dev Tools & Software (VS Code, Postman, GitHub, GIS software licenses, testing tools) | Free |
| Miscellaneous (Travel to plantations, field trials, calibration, replacements) | 10,000 |
| Total Estimated Budget | 130,000 |

## 12. COCLUSION

The proposed SMARTCINNAMON framework introduces an innovative, fully integrated digital decision-support system designed to address persistent challenges in cinnamon cultivation, post-harvest quality assurance, land utilization, and disease management. By combining IoT, GIS, AI, and machine learning, this solution shifts from traditional experience driven practices toward data-driven, automated, and predictive agriculture.

The four specialized components deliver end-to-end support across the cinnamon value chain:

- The Cinnamon-Specific Fertilizer Deficit System enables precision nutrient management through IoT sensor data and ML-driven fertilizer recommendations, reducing overuse and underuse of inputs.
- The Post-Harvest Cinnamon Quality Guard safeguards storage quality by monitoring ten environmental parameters and applying ML-based risk detection to prevent mold growth and aroma degradation, ensuring HACCP and GI compliance.
- The Cinnamon GIS Yield Planner leverages drone/satellite imagery and ML segmentation to optimize replanting, calculate available land, and forecast yields, enabling proactive plantation planning without costly surveys.
- The Mobile-Based Fungal Detection & Spread Predictor transforms traditional visual inspections into quantitative diagnostics by detecting Cylindrocladium and Anthracnose, calculating severity, and predicting future spread for timely interventions.

Collectively, these modules form a holistic ecosystem that enhances productivity, quality, and sustainability for Sri Lanka's cinnamon industry. The framework not only empowers farmers with real-time insights and actionable recommendations but also strengthens Sri Lanka's export competitiveness by ensuring consistency, compliance, and premium quality standards.

The project demonstrates how smart agriculture technologies can modernize cinnamon cultivation from roots to export, providing a scalable and affordable blueprint for digital transformation in spice industries. Future research can extend the system with AI-driven predictive analytics for climate risks, blockchain-enabled supply chain traceability, and farmer-cooperative integration, further reinforcing Sri Lanka's leadership in the global cinnamon market.

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# 14. APPENDICIES

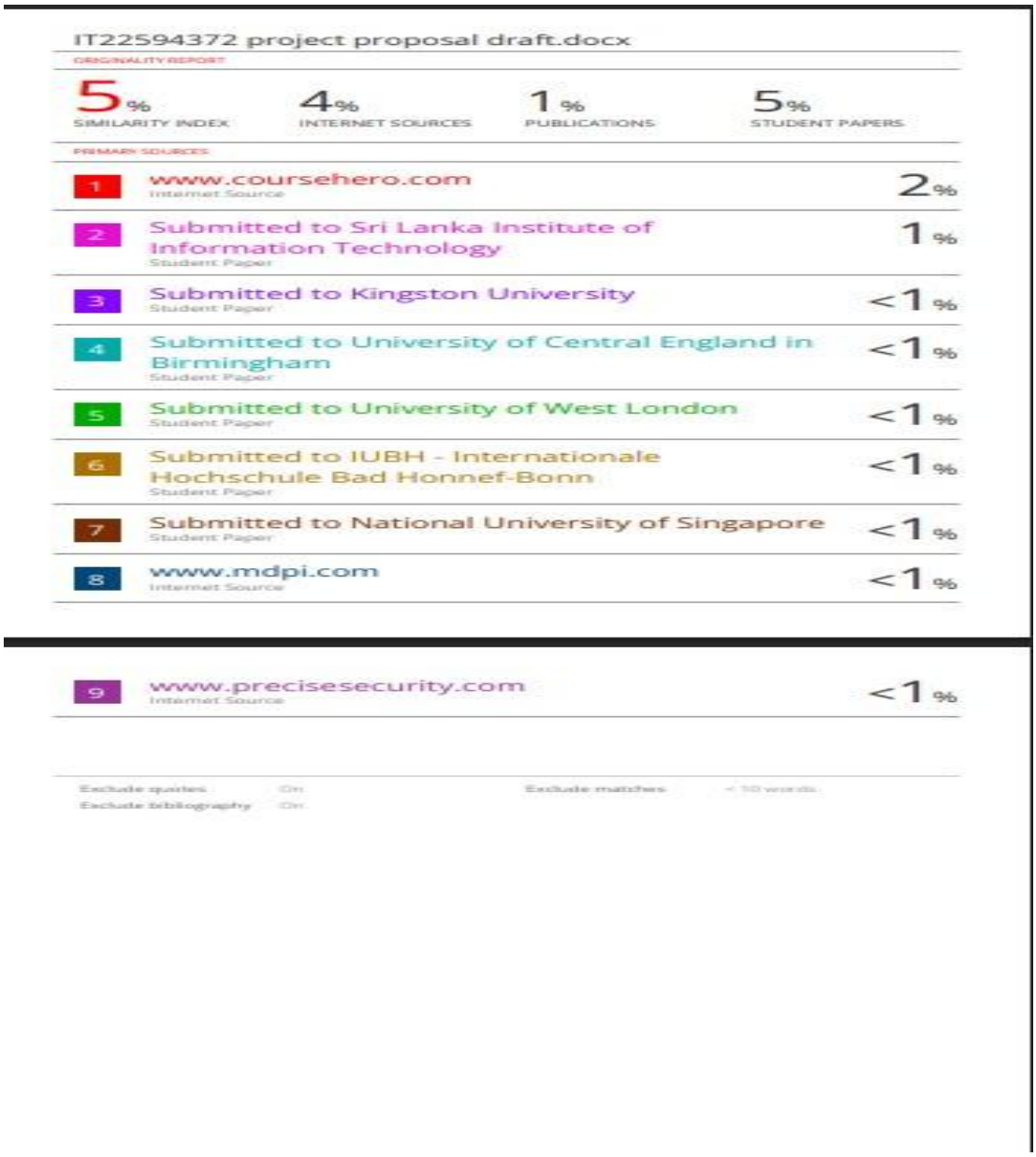

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*Figure 14.1: Appendices I*